

High-level summary of derivations in these lectures

This set of notes focuses on the hydrodynamic response of electron fluids at low and high frequencies. We already know that in the static, long wavelength limit, the dielectric function is determined by the compressibility (see **Appendix 1** and **Appendix 2** below). This simply follows from the notion that in the static limit forces must be balanced (and the only forces at play are the pressure of the nearly free electron gas and the electrostatic interactions between the electrons).

Starting in **Section 1**, below, we go beyond the strictly static limit. We briefly describe the main results we derive: In **Section 1**, we introduce the main equation which describes nonlocal electronic response and show how the free parameter, β^2 , of this equation may be found through a series expansion of the plasma frequency. In addition, we show how the nonlocality of the electronic response changes the longitudinal conductivity and the longitudinal dielectric function. In **Section 2(a, b, c)**, we calculate, from first principles, the effect of nonlocality for a 3D electron liquid, a 2D electron liquid, and a linearly dispersing band in two dimensions, using the random phase approximation. By doing so, we find nonlocal corrections to the plasma frequency and the conductivity, thus obtaining β^2 . In **Section 3**, we derive the hydrodynamic equation from classical arguments and we show why β^2 found from the classical formulation does not match β^2 found in the RPA (RPA shows β^2 must be $3v_F^2/5$ in 3D and $3v_F^2/4$ in 2D but, for instance, classical arguments seem to say $\beta^2 = v_F^2/3$ in 3D). In **Sections 4 and 5**, we determine β^2 for graphene in the hydrodynamic approach as well as within the RPA (however, in this case, we look at the RPA in a two-band approach, in contrast with the one-band approach in **Section 2c**). The hydrodynamic approach in **Section 4** gives us the expression for pressure relevant for **Appendix 2**. However, β^2 derived in this way is $v_F^2/2$ in contrast with $3v_F^2/4$ in the one-band RPA. The interpretation of the two-band RPA from a classical perspective is not altogether clear since one obtains two different β^2 parameters. In addition, β^2 varies strongly with frequency. The value $3v_F^2/4$ is only found for the "intraband" β^2 at frequencies below the Fermi energy.

Section 1: Hydrodynamic response formulation

In this section, we simply state the hydrodynamic equation (see, e.g. [Unusual resonances in nanoplasmonic structures due to nonlocal response](#)

([https://journals.aps.org/prb/pdf/10.1103/PhysRevB.84.121412?](https://journals.aps.org/prb/pdf/10.1103/PhysRevB.84.121412?casa_token=LI3ua9Xq_tsAAAAA%3AU2TYIB471bO0JwfQ5mgmTD9A_FgWQBaRiu2ENBTQ0YHJfw3c_Vg-3fwBm-J93HGopKh3ZalVHAPMcQ)

[casa_token=LI3ua9Xq_tsAAAAA%3AU2TYIB471bO0JwfQ5mgmTD9A_FgWQBaRiu2ENBTQ0YHJfw3c_Vg-3fwBm-J93HGopKh3ZalVHAPMcQ](https://journals.aps.org/prb/pdf/10.1103/PhysRevB.84.121412?casa_token=LI3ua9Xq_tsAAAAA%3AU2TYIB471bO0JwfQ5mgmTD9A_FgWQBaRiu2ENBTQ0YHJfw3c_Vg-3fwBm-J93HGopKh3ZalVHAPMcQ)); we derive it in **Section 3**):

$$\mathbf{K}(\mathbf{r}, \omega) + \frac{\beta^2}{\omega^2} \nabla(\nabla \cdot \mathbf{K}(\mathbf{r}, \omega)) = \sigma(\omega) \mathbf{E}(\mathbf{r}, \omega),$$

where β has units of velocity. In the $\beta \rightarrow 0$ limit, we have a purely local response, $\mathbf{K}(\mathbf{r}, \omega) = \sigma(\omega) \mathbf{E}(\mathbf{r}, \omega)$, which relates $\mathbf{K}(\mathbf{r}, \omega)$ —the current density (either per unit area or per unit length depending on whether we are in two or three dimensions)—to the electric field, $\mathbf{E}(\mathbf{r}, \omega)$ at the same physical location. The introduction of the parameter β^2 makes the current-field relation nonlocal, so it is instructive to look at this relation in momentum space instead, in which it reads:

$$\mathbf{K}(\mathbf{q}, \omega) - \frac{\beta^2}{\omega^2} \mathbf{q}(\mathbf{q} \cdot \mathbf{K}(\mathbf{q}, \omega)) = \sigma(\omega) \mathbf{E}(\mathbf{q}, \omega)$$

Clearly, the transverse, i.e. $\mathbf{q} \perp \mathbf{E}, \mathbf{K}$, response is unaffected and the spatially local current-field relation still holds for the transverse response. The longitudinal conductivity and longitudinal dielectric functions are altered, however, and are given by (**for three dimensions**):

$$\sigma_L(\mathbf{q}, \omega) = \frac{\sigma(\omega)}{1 - \beta^2 |\mathbf{q}|^2 / \omega^2} = \frac{ine^2/m}{\omega - \beta^2 |\mathbf{q}|^2 / \omega}$$

$$\varepsilon_L(\mathbf{q}, \omega) = 1 - \frac{\omega_p^2}{\omega^2 - \beta^2 |\mathbf{q}|^2} \rightarrow \omega_p^2(\mathbf{q}) = \omega_p^2 + \beta^2 |\mathbf{q}|^2,$$

where we have introduced the local bulk plasmon frequency, $\omega_p^2 = ne^2/m\varepsilon_0$. The above expression for the nonlocal plasmon frequency is exact in the hydrodynamic approximation but will arise as the term proportional to $|\mathbf{q}|^2$ in a series expansion in the RPA.

In two dimensions, we have, instead,

$$\varepsilon_L(\mathbf{q}, \omega) = 1 - \frac{ne^2|\mathbf{q}|/2m\varepsilon_0}{\omega^2 - \beta^2|\mathbf{q}|^2} \rightarrow \omega_p^2(\mathbf{q}) = \omega_p^2 + \beta^2|\mathbf{q}|^2,$$

where $\omega_p^2 = ne^2m|\mathbf{q}|/2\varepsilon_0$ is the plasmon dispersion in 2d without nonlocal corrections. Therefore, the effect of β^2 is similar for both cases.

To find the nonlocal corrections, it suffices, then, to compute the dispersion of the plasmon mode to order $|\mathbf{q}|^2$ or to, order-by-order, match terms in the conductivity (or dielectric function).

Section 2a: Derivation of β for electron gas in 3 dimensions

As a warmup, we calculate β^2 for the free electron gas. We do this by calculating the correction to the bulk plasmon frequency,

We take the standard Lindhard function, see e.g. [Lindhard Theory](https://en.wikipedia.org/wiki/Lindhard_theory)

(https://en.wikipedia.org/wiki/Lindhard_theory). (note we do not include spin degeneracy in the Lindhard function itself and introduce it afterwards) and take $-\mathbf{k}' = \mathbf{k} + \mathbf{q}$ to transform it into:

$$\sum_{\mathbf{k}} \frac{f_{\mathbf{k}}}{\hbar\omega + \varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}+\mathbf{q}}} - \frac{f_{\mathbf{k}}}{\hbar\omega + \varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\mathbf{k}}} = \sum_{\mathbf{k}} \frac{2f_{\mathbf{k}}(\varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\mathbf{k}})}{(\hbar\omega)^2 - (\varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\mathbf{k}})^2} =$$

$$\frac{\hbar^2}{m} \sum_{\mathbf{k} < \mathbf{k}_F} \frac{(|\mathbf{q}|^2 + 2\mathbf{k} \cdot \mathbf{q})}{(\hbar\omega)^2 - (\hbar^2/2m)^2(|\mathbf{q}|^2 + 2\mathbf{k} \cdot \mathbf{q})^2}$$

At high frequencies, the lowest order term will be given by (with g_s denoting degeneracy of quantum numbers, i.e. spin and Ω being volume):

$$\varepsilon(\mathbf{q}, \omega) = 1 - e^2 g_s \chi^1 / (\Omega \varepsilon_0 |\mathbf{q}|^2) = 1 - \frac{e^2 g_s}{m \varepsilon_0} \left(\frac{4\pi k_F^3}{3(8\pi^3)\omega^2} \right) = 1 - \frac{ne^2}{m \varepsilon_0 \omega^2} = 1 - \frac{\omega_p^2}{\omega^2},$$

where we used the fact that the electron density, n , is given by:

$$n = \frac{4\pi g_s k_F^3}{3(2\pi)^3} = \frac{k_F^3}{3\pi^2}.$$

The next order is given by:

$$\begin{aligned} \chi^2 &= \frac{\hbar^2}{m(\hbar\omega)^4} \left[\frac{\hbar^2}{2m} \right]^2 \sum_{\mathbf{k} < \mathbf{k}_F} \left[|\mathbf{q}|^2 + 2\mathbf{k} \cdot \mathbf{q} \right]^3 = \\ &= \frac{\hbar^6 \Omega}{4m^3 (\hbar\omega)^4 (4\pi^2)} \int \left[q^2 + 2kqu \right]^3 k^2 du dk \rightarrow \\ &= \frac{(3 \times 4) \hbar^6 \Omega |\mathbf{q}|^4}{4m^3 (\hbar\omega)^4 (4\pi^2)} \int k^4 u^2 du dk = \frac{2}{3} \frac{k_F^5}{5} \frac{3\hbar^2 \Omega |\mathbf{q}|^4}{m^3 \omega^4 (4\pi^2)} = \\ &= \frac{3}{5} \frac{k_F^3 \Omega}{3\pi^2} \frac{1}{g_s m} |\mathbf{q}|^4 \left[\frac{\hbar k_F}{m} \right]^2 \rightarrow \varepsilon(\mathbf{q}, \omega) = 1 - \frac{\omega_p^2}{\omega^2} - \frac{3v_F^2 q^2}{5} \frac{\omega_p^2}{\omega^4} \end{aligned}$$

In the first equality, we used the fact that we may, wlog, put $\mathbf{q}$ along the e_z direction and integrate in the azimuthal direction. Furthermore, we use the fact that the only terms that we consider are proportional to $|\mathbf{q}|^4$. If we define $\beta^2 = 3v_F^2/5$, we will have:

$$\begin{aligned} \varepsilon(\mathbf{q}, \omega) &= 1 - \frac{\omega_p^2}{\omega^2} - \frac{\omega_p^2 \beta^2 q^2}{\omega^4} \rightarrow \omega_p^2(\mathbf{q}) = \omega_p^2/2 \pm \sqrt{\omega_p^4 + 4\omega_p^2 \beta^2 q^2}/2 \approx \\ &= \frac{\omega_p^2}{2} \pm \frac{\omega_p^2}{2} \left[1 + \frac{2\beta^2 q^2}{\omega_p^2} \right] \rightarrow \omega_p^2(\mathbf{q}) = \omega_p^2 + \beta^2 |\mathbf{q}|^2, \end{aligned}$$

where the last expression we derived for the nonlocal plasmon dispersion demonstrates that the β derived in this section is the same as the β which corresponds to the hydrodynamic current-field relation. In addition to the nonlocal longitudinal permittivity, we may also write out the longitudinal conductivity as:

$$\sigma(\mathbf{q}, \omega) = \frac{i\varepsilon_0 \omega_p^2}{\omega} \left[1 + \frac{|\mathbf{q}|^2}{\omega^2} \beta^2 \right]$$

Section 2b: Derivation of β for electron gas in 2 dimensions

As in the previous subsection, we take the standard Lindhard function and take $-\mathbf{k}' = \mathbf{k} + \mathbf{q}$ to transform it into:

$$\sum_{\mathbf{k}} \frac{f_{\mathbf{k}}}{\hbar\omega + \varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}+\mathbf{q}}} - \frac{f_{\mathbf{k}}}{\hbar\omega + \varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\mathbf{k}}} = \sum_{\mathbf{k}} \frac{2f_{\mathbf{k}}(\varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\mathbf{k}})}{(\hbar\omega)^2 - (\varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\mathbf{k}})^2} =$$

$$\frac{\hbar^2}{m} \sum_{\mathbf{k} < \mathbf{k}_F} \frac{(|\mathbf{q}|^2 + 2\mathbf{k} \cdot \mathbf{q})}{(\hbar\omega)^2 - (\hbar^2/2m)^2(|\mathbf{q}|^2 + 2\mathbf{k} \cdot \mathbf{q})^2}$$

The lowest order term of the Lindhard function is, again,

$$\chi^1(\mathbf{q}, \omega) = \frac{N|\mathbf{q}|^2}{m\omega^2} \rightarrow \varepsilon(\mathbf{q}, \omega) = 1 - \frac{ne^2|\mathbf{q}|}{2m\varepsilon_0\omega^2}$$

The next order is given by:

$$\chi^2(\mathbf{q}, \omega) = \frac{\hbar^2}{m(\hbar\omega)^4} \left[\frac{\hbar^2}{2m} \right]^2 \sum_{\mathbf{k} < \mathbf{k}_F} [|\mathbf{q}|^2 + 2\mathbf{k} \cdot \mathbf{q}]^3 =$$

$$\frac{\hbar^6 \Omega}{4m^3(\hbar\omega)^4(4\pi^2)} \int \left[q^2 + 2kq \cos(\theta) \right]^3 k d\theta dk \rightarrow$$

$$\frac{\hbar^2 \Omega (3 \times 4) |\mathbf{q}|^4}{4m^3 \omega^4 (4\pi^2)} \int k^3 \cos^2(\theta) dk d\theta = \frac{3\hbar^2 \Omega |\mathbf{q}|^4 k_F^4}{4m^3 \omega^4 (4\pi)} = \frac{3}{4} \left[\frac{\hbar k_F}{m} \right]^2 \frac{N|\mathbf{q}|^4}{mg_s \omega^4}$$

From this, we obtain the dielectric function:

$$\varepsilon(\mathbf{q}, \omega) = 1 - \frac{\omega_p^2}{\omega^2} - \frac{3v_F^2 |\mathbf{q}|^2 \omega_p^2}{4\omega^4}$$

As well as the conductivity,

$$\sigma(\mathbf{q}, \omega) = \frac{2i\varepsilon_0\omega_p^2}{|\mathbf{q}|\omega} \left[1 + \frac{3v_F^2}{4} \frac{|\mathbf{q}|^2}{\omega^2} \right]$$

Section 2c: Derivation of β for a linear dispersion in 2 dimensions

As in the previous subsection, we take the standard Lindhard function and take $-\mathbf{k}' = \mathbf{k} + \mathbf{q}$ to transform it into:

$$\sum_{\mathbf{k}} \frac{2f_{\mathbf{k}}(\varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\mathbf{k}})}{(\hbar\omega)^2 - (\varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\mathbf{k}})^2} \approx (2\hbar v_F) \sum_{\mathbf{k} < \mathbf{k}_F} \frac{\sqrt{|\mathbf{k}|^2 + |\mathbf{q}|^2 + 2\mathbf{k} \cdot \mathbf{q}} - |\mathbf{k}|}{(\hbar\omega)^2}$$

We expand $\sqrt{|\mathbf{k}|^2 + |\mathbf{q}|^2 + 2\mathbf{k} \cdot \mathbf{q}} \approx |\mathbf{k}| \sqrt{1 + |\mathbf{q}|^2/|\mathbf{k}|^2 + 2\mathbf{k} \cdot \mathbf{q}/|\mathbf{k}|^2}$

We need to isolate all $|\mathbf{q}|^2$ terms:

$$\frac{|\mathbf{q}|^2}{2|\mathbf{k}|} - \frac{4(\mathbf{k} \cdot \mathbf{q})^2}{8|\mathbf{k}|^3} = \frac{|\mathbf{q}|^2}{2} \left[\frac{1 - \cos^2(\theta)}{k} \right]$$

Therefore, we write the lowest order term of the Lindhard function as (using the fact that $\sin^2(\theta)$ will give us a factor of π):

$$\chi^1 = \frac{v_F |\mathbf{q}|^2 \Omega k_F}{\hbar \omega^2 (4\pi)} = \frac{|\mathbf{q}|^2 \Omega k_F^2}{m (4\pi) \omega^2} \rightarrow \varepsilon(\mathbf{q}, \omega) = 1 - \frac{ne^2 |\mathbf{q}|}{2m\varepsilon_0 \omega^2}, \sigma(\mathbf{q}, \omega) = \frac{ine^2}{m\omega}$$

where we introduced the effective mass $m = \hbar k_F / v_F$. We look now into the second order term:

$$\frac{2\hbar^3 v_F^3}{(\hbar\omega)^4} \sum_{\mathbf{k}} \left[|\mathbf{k}| \sqrt{1 + |\mathbf{q}|^2/|\mathbf{k}|^2 + 2\mathbf{k} \cdot \mathbf{q}/|\mathbf{k}|^2} - |\mathbf{k}| \right]^3$$

Within the bracket, we isolate the terms proportional to $|\mathbf{q}|^4$:

$$\left[\frac{1}{2} \frac{|\mathbf{q}|^2 \sin^2(\theta)}{|\mathbf{k}|} + \frac{kq \cos(\theta)}{|\mathbf{k}|} \right]^3 \rightarrow \frac{3|\mathbf{q}|^4}{2k} \sin^2(\theta) \cos^2(\theta)$$

So we obtain (using the fact that $\sin^2(\theta) \cos^2(\theta) = \sin^2(2\theta)/4$):

$$\chi^2(\mathbf{q}, \omega) = \frac{3v_F^3 |\mathbf{q}|^4 k_F^2 \Omega}{4\hbar k_F \omega^4 (4\pi)} = \frac{3v_F^2}{4} \left[\frac{N}{g_s g_v m} \frac{|\mathbf{q}|^4}{\omega^4} \right]$$

$$\rightarrow \varepsilon(\mathbf{q}, \omega) \approx 1 - \frac{ne^2 |\mathbf{q}|}{2m\varepsilon_0 \omega^2} - \frac{3v_F^2}{4} \frac{e^2 n |\mathbf{q}|^3}{2m\varepsilon_0 \omega^4} = 1 - \frac{\omega_p^2}{\omega^2} \left[1 + \frac{3v_F^2}{4} \frac{|\mathbf{q}|^2}{\omega^2} \right]$$

Aside: Comparison to Marinko's paper

We compare the results in the last subsection with those reported in Plasma echoes in graphene ([https://journals.aps.org/prb/pdf/10.1103/PhysRevB.108.085404?](https://journals.aps.org/prb/pdf/10.1103/PhysRevB.108.085404?casa_token=lu4cXI8WMmAAAAAA%3AANPgfV5oppL0fnSYtQstl3p-Sztr0tHxHWDJKuD45le5PYeAf0KLQngBwjXrs2z9HAEPZgoJPtok)

[casa_token=lu4cXI8WMmAAAAAA%3AANPgfV5oppL0fnSYtQstl3p-](https://journals.aps.org/prb/pdf/10.1103/PhysRevB.108.085404?casa_token=lu4cXI8WMmAAAAAA%3AANPgfV5oppL0fnSYtQstl3p-Sztr0tHxHWDJKuD45le5PYeAf0KLQngBwjXrs2z9HAEPZgoJPtok)

[Sztr0tHxHWDJKuD45le5PYeAf0KLQngBwjXrs2z9HAEPZgoJPtok](https://journals.aps.org/prb/pdf/10.1103/PhysRevB.108.085404?casa_token=lu4cXI8WMmAAAAAA%3AANPgfV5oppL0fnSYtQstl3p-Sztr0tHxHWDJKuD45le5PYeAf0KLQngBwjXrs2z9HAEPZgoJPtok)), **Equation 16** (where we also use the notation of that paper in this aside):

$$1 + \chi_k \left(1 - \frac{1}{\sqrt{1 - k^2 v_F^2 / \omega^2}} \right) \approx 1 - \chi_k \frac{k^2 v_F^2}{2\omega^2}$$

We deal only with low temperatures, so we expand χ_k in the low temperature regime:

$$\chi_k = \frac{e^2 k_B T}{k\varepsilon_0 \pi \hbar^2 v_F^2} \ln \left[1 + e^{\varepsilon_F / k_B T} \right] \approx \frac{e^2 \varepsilon_F}{k\varepsilon_0 \pi \hbar^2 v_F^2} \rightarrow \varepsilon \approx 1 - \frac{e^2 v_F k_F^2 k}{2\varepsilon_0 \pi k_F \hbar \omega^2}$$

We write

$$k_F^2 / \pi = n \rightarrow \varepsilon \approx 1 - \frac{ne^2 k}{2m\varepsilon_0 \omega^2},$$

where we used our expression for the graphene effective mass. Encouraged by the fact that the lowest order expansion matches, we check the next order term:

$$\varepsilon \approx 1 + \chi_k \left(-\frac{k^2 v_F^2}{2\omega^2} - \frac{3k^4 v_F^4}{8\omega^4} \right) = 1 - \frac{ne^2 k}{2m\varepsilon_0 \omega^2} \left[1 + \frac{3v_F^2}{4} \frac{k^2}{\omega^2} \right]$$

So, again, we obtain $\beta^2 = 3v_F^2/4$

Section 3: Derivation from hydrodynamic pressure

At this point, we point out an interesting fact, following Electrodynamics of a Layered Electron Gas. I. Single Layer (<https://www.sciencedirect.com/science/article/pii/0003491673901619>), by **Alexander Fetter** as well as Electromagnetic Surface Modes (<https://www.google.com/url?sa=t&source=web&rct=j&opi=899784449&url=https://ia601409.us.archive.org/12/items/in.ernet.dli.2015.148030/2015.148030.Electromagnetic-Surface-Modes.pdf&ved=2ahUKEwj-t4Pvs9aPAXXkD1kFHZ6oFk0QFnoECB4QAQ&usg=AOvVaw3IVFcvFjy3p2Me4T824Kv>), by **Boardman**. In particular, we try to derive the correction to the plasma frequency above through a hydrodynamic approach, in the vein of Unusual resonances in nanoplasmonic structures due to nonlocal response (https://journals.aps.org/prb/pdf/10.1103/PhysRevB.84.121412?casa_token=LI3ua9Xq_tsAAAAA%3AU2TYIB471bO0JwfQ5mgmTD9A_FgWQBaRiu2ENBTQ0YHJfw3c_Vg-3fwBm-J93HGopKh3ZalVHAPMcQ). In this approach, we write the equation of motion, $\partial_t \mathbf{v} = -e\mathbf{E}/m - [P(\mathbf{x} + \mathbf{dx}) - P(\mathbf{x})]A/(mnA|\mathbf{dx}|)$, where the numerator, the total force acting on a thin slab of the material and $nA\mathbf{dx}$ is the total number of electrons within the slab. We may therefore rewrite the equation of motion as:

$$\partial_t \mathbf{v} = -e\mathbf{E}/m - \frac{1}{nm} \frac{\partial P}{\partial n} \nabla n$$

The energy of a free electron gas is given by $\frac{3}{5} E_F N$, which means that the pressure is given by

$$P = \frac{2}{5} E_F n \rightarrow \frac{1}{m} \frac{\partial P}{\partial n} = \frac{1}{3} \frac{\hbar^2 k_F^2}{m^2} = \frac{v_F^2}{3}. \text{ Putting this all together, we obtain:}$$

$$\partial_t \mathbf{v} = -\frac{e}{m} \mathbf{E} - \frac{v_F^2}{3n} \nabla n$$

We may now write this in terms of the current, $-env$, to obtain:

$$\partial_t \mathbf{j} = \varepsilon_0 \omega_p^2 \mathbf{E} + \frac{ev_F^2}{3} \nabla n$$

We may now use the continuity equation, $\nabla \cdot \mathbf{j} = e\partial_t n$ and take a second time derivative on both sides to obtain:

$$(-i\omega)^2 \mathbf{j} = \varepsilon_0 \omega_p^2 (-i\omega) \mathbf{E} + \frac{v_F^2}{3} \nabla(\nabla \cdot \mathbf{j}) \rightarrow \mathbf{j} + \frac{v_F^2}{3\omega^2} \nabla(\nabla \cdot \mathbf{j}) = \sigma(\omega) \mathbf{E},$$

which is the equation at the top of this document with $\beta^2 = v_F^2/3$. However, here comes the rub: the RPA calculation in the previous section demonstrates conclusively that we must have, instead, $\beta^2 = 3v_F^2/5$. The paper by **Fetter** mentions that this discrepancy is due to the high frequency (not static) expansion, though he does not provide a derivation of the correct prefactor and simply mentions that the dimensionally-dependent correction factor is given by $3D/(D+2)$, which is equal to $9/5$ in three dimensions, $3/2$ in two dimensions, and 1 in one dimension. Let's now see how this correction factor may come about.

An aside: Equation 10.99 in Jackson E&M

We wish to verify **Equation 10.99** in Jackson, which relates the average square velocity to the Fermi velocity squared (in **three spatial dimensions**):

$$\frac{1}{N} \sum_{\mathbf{k}} |v_{\mathbf{k},i}|^2 = \frac{1}{n} \frac{1}{2\pi^2} \int \left(\frac{\hbar k}{m} \right)^2 k^2 \cos^2(\theta) \sin(\theta) d\theta dk = \frac{1}{n} \frac{k_F^3}{3\pi^2} \frac{1}{5} \left[\frac{\hbar k_F}{m} \right]^2 = \frac{v_F^2}{5}$$

QED.

Let's see how different this is from the answer in one dimension,

$$\frac{2L}{N\pi} \left[\frac{k_F}{3} \right] \left[\frac{\hbar k_F}{m} \right]^2 = \frac{v_F^2}{3}$$

And in two dimensions:

$$\frac{1}{n2\pi^2} \left(\frac{\hbar}{m} \right)^2 \int k^3 \cos^2(\theta) d\theta dk = \frac{k_F^2}{8n\pi} \left(\frac{\hbar k_F}{m} \right)^2 = \frac{v_F^2}{4}$$

The reason we derive these expressions is that at high frequencies pressure waves become essentially one dimensional, so we need to calculate the pressure in one dimension and express it in terms of the Fermi velocity of the three dimensional system.

Pressure in one dimension

The energy per particle in one dimension for free electrons is given by (taking into account one factor of 2 for spin degeneracy and another factor of 2 to reduce the integration range from $-k_F < k < k_F$ to $0 < k < k_F$):

$$\frac{2 \times 2 \times \Omega}{(2\pi)N} \frac{\hbar^2}{2m} \int_0^{k_F} k^2 dk = \frac{E_F}{3} \frac{2\Omega k_F}{\pi N},$$

where

$$N = \frac{2k_F\Omega}{\pi} \rightarrow E/N = \frac{E_F}{3} \rightarrow P = \frac{2E_F}{3}n \propto n^3 \rightarrow \frac{1}{m} \frac{\partial P}{\partial n} = \frac{\hbar^2 k_F^2}{m^2} = 3\langle u^2 \rangle = \frac{3}{5}v_{F,3D}^2,$$

where we used the results in the previous section to express the Fermi momentum squared in one dimension in terms of the average square velocity in one dimension. We then expressed the average square velocity in one dimension in terms of the three dimensional Fermi velocity.

In two dimensions, the energy per particle for free electrons is given by:

$$\frac{2\Omega}{(4\pi^2)N} \frac{\hbar^2}{2m} (2\pi) \frac{k_F^4}{4} = \frac{E_F}{N} \frac{\Omega k_F^2}{4\pi},$$

where

$$N = \frac{k_F^2\Omega}{2\pi} \rightarrow E/N = \frac{E_F}{2} \rightarrow P = \frac{E_F n}{2} \rightarrow \frac{1}{m} \frac{\partial P}{\partial n} = \frac{\hbar^2 k_F^2}{2m^2} = v_F^2/2$$

Thus, just as the naive approach gave us $\beta^2 = v_F^2/3$ in 3D, we get $\beta^2 = v_F^2/2$ in 2D. Applying the Fetter correction factor, $3/2$, gives us the correct answer of $3v_F^2/4$, which is consistent with the RPA approach. Equivalently, if we had considered the pressure waves as being one dimensional, we would have also gotten $\beta^2 = \frac{3}{4}v_{F,2D}^2$.

Section 4: Hydrodynamic corrections for graphene

For graphene, we have that the energy is given by (taking into account valley and spin degeneracy):

$$\frac{\Omega}{\pi^2} (2\pi) \int_0^{k_F} \hbar v_F k^2 dk = \frac{2\Omega}{3\pi} \hbar v_F k_F^3$$

The total number of electrons is given by:

$$N = \frac{\pi k_F^2 \Omega}{\pi^2} = \Omega \frac{k_F^2}{\pi}$$

Therefore, we may write:

$$E = \frac{2}{3} (\hbar v_F k_F) N.$$

Let's first find the pressure the normal way in two spatial dimensions, noting that $k_F \propto 1/\sqrt{\Omega} \rightarrow P = \frac{1}{3} (\hbar v_F k_F) n$. Taking the derivative of this with respect to n then gives us $\frac{1}{2} \hbar v_F k_F \rightarrow \beta^2 = \frac{v_F^2}{2}$ (where we used the effective mass $m = \hbar k_F / v_F$ derived in the next section). Clearly, our answer is off by a factor of $3/2$, due to the high frequency expansion. Our result is consistent with **equation 2.35** in the following Master's thesis:

<https://www.proquest.com/docview/2917520241?fromopenview=true&pq-origsite=gscholar&sourcetype=Dissertations & Theses>
(<https://www.proquest.com/docview/2917520241?fromopenview=true&pq-origsite=gscholar&sourcetype=Dissertations%20&%20Theses>).

Section 5: Pressure corrections in the RPA for graphene

In this section, we refer to two relevant papers: Dynamical polarization of graphene at finite doping (<https://iopscience.iop.org/article/10.1088/1367-2630/8/12/318/pdf>), as well as the supplemental information in the paper:

Classical and Quantum Plasmonics in Graphene Nanodisks:

the Role of Edge States ([https://link.aps.org/pdf/10.1103/PhysRevB.90.241414?](https://link.aps.org/pdf/10.1103/PhysRevB.90.241414?casa_token=GwqXhy0kCv0AAAAA:C1i4xZyQE8U6sSKjdi09B3rh_8K0unbxdCOWhalyEgpAzlQblvZ9I2nNmsKJxudWAZDr_oHfZi3uBclg)

[casa_token=GwqXhy0kCv0AAAAA:C1i4xZyQE8U6sSKjdi09B3rh_8K0unbxdCOWhalyEgpAzlQblvZ9I2nNmsKJxudWAZDr_oHfZi3uBclg](https://link.aps.org/pdf/10.1103/PhysRevB.90.241414?casa_token=GwqXhy0kCv0AAAAA:C1i4xZyQE8U6sSKjdi09B3rh_8K0unbxdCOWhalyEgpAzlQblvZ9I2nNmsKJxudWAZDr_oHfZi3uBclg)).

We introduce non-dimensionalized quantities, $\tilde{k} = k/k_F$, $\tilde{\omega} = \hbar\omega/\varepsilon_F$, and we write the conductivity at arbitrary wavevector and frequency as (**equation S23a** in the second reference):

$$\frac{ie^2\tilde{\omega}}{\pi\hbar\tilde{k}^2} \left[-2 + \frac{1}{4}F(\tilde{k}, \tilde{\omega}) \left[W\left(\frac{2+\tilde{\omega}}{\tilde{k}}\right) - W\left(\frac{2-\tilde{\omega}}{\tilde{k}}\right) \right] \right],$$

where **equation S23b** gives us the functions F, W :

$$W = x\sqrt{x^2 - 1} - \ln(x + \sqrt{x^2 - 1}), F(\tilde{k}, \tilde{\omega}) = \frac{\tilde{k}^2}{\sqrt{\tilde{\omega}^2 - \tilde{k}^2}}$$

We investigate the behavior in terms of powers of the momentum:

$$\frac{\tilde{k}^2}{\sqrt{\tilde{\omega}^2 - \tilde{k}^2}} \approx \frac{\tilde{k}^2}{\tilde{\omega}} \left[1 + \frac{\tilde{k}^2}{2\tilde{\omega}^2} + \frac{3\tilde{k}^4}{8\tilde{\omega}^4} \right]$$

$$x\sqrt{x^2 - 1} = x^2\sqrt{1 - 1/x^2} \approx \left(x^2 - \frac{1}{2} - \frac{1}{8x^2} \right),$$

where we also used the fact that $x > 0$ in the frequency regime of interest.

Therefore, we have that

$$\begin{aligned} \frac{1}{4} F(\tilde{k}, \tilde{\omega}) \left[W\left(\frac{2+\tilde{\omega}}{\tilde{k}}\right) - W\left(\frac{2-\tilde{\omega}}{\tilde{k}}\right) \right] &\approx \frac{1}{4\tilde{\omega}} \left[1 + \frac{\tilde{k}^2}{2\tilde{\omega}^2} + \frac{3\tilde{k}^4}{8\tilde{\omega}^4} \right] \times \\ &\left[(2+\tilde{\omega})^2 - (2-\tilde{\omega})^2 - \frac{\tilde{k}^4}{8(2+\tilde{\omega})^2} + \frac{\tilde{k}^4}{8(2-\tilde{\omega})^2} \right] = \\ &\left[2 + \frac{\tilde{k}^2}{\tilde{\omega}^2} + \frac{3\tilde{k}^4}{4\tilde{\omega}^4} + \frac{\tilde{k}^4[(2+\tilde{\omega})^2 - (2-\tilde{\omega})^2]}{32\tilde{\omega}(4-\tilde{\omega}^2)^2} \right] = \\ &2 + \frac{\tilde{k}^2}{\tilde{\omega}^2} + \frac{3\tilde{k}^4}{4\tilde{\omega}^4} + \frac{\tilde{k}^4}{4\tilde{\omega}^4(4/\tilde{\omega}^2 - 1)^2} \end{aligned}$$

Which gives us the conductivity in powers of the wavevector:

$$\frac{ie^2\tilde{\omega}}{\pi\hbar\tilde{k}^2} \left[\frac{\tilde{k}^2}{\tilde{\omega}^2} + \frac{3\tilde{k}^4}{4\tilde{\omega}^4} + \frac{\tilde{k}^4}{4\tilde{\omega}^4(4/\tilde{\omega}^2 - 1)^2} \right] = \frac{ie^2\varepsilon_F}{\pi\hbar^2\omega} \left[1 + \frac{3}{4} \frac{\tilde{k}^2}{\tilde{\omega}^2} + \frac{\tilde{k}^2}{4\tilde{\omega}^2(4/\tilde{\omega}^2 - 1)^2} \right]$$

Note that

$$\frac{\tilde{k}^2}{\tilde{\omega}^2} = \frac{k^2\varepsilon_F^2}{\hbar^2\omega^2k_F^2} = \frac{k^2}{\omega^2}v_F^2$$

Therefore, we have that in graphene the nonlocality is controlled by $\beta^2 = 3v_F^2/4$. We may also read off the graphene effective mass from the local conductivity. Since, if we note that

$$\frac{ine^2}{m\omega} = \frac{ie^2\varepsilon_F}{\pi\hbar^2\omega} \rightarrow \frac{n}{m} = \frac{\varepsilon_F}{\pi\hbar^2} \rightarrow m = \frac{k_F^2}{\pi} \frac{\pi\hbar^2}{\varepsilon_F} = \frac{\hbar k_F}{v_F}$$

Derivation of β_{inter}^2

The logarithmic contribution to the conductivity will be given by (isolating the logarithmic contribution from the function $W(x)$):

$$\frac{ie^2\tilde{\omega}}{\pi\hbar\tilde{k}^2} \frac{\tilde{k}^2}{4\tilde{\omega}} \left[1 + \frac{\tilde{k}^2}{2\tilde{\omega}^2} + \frac{3\tilde{k}^4}{8\tilde{\omega}^4} \right] \times \ln \left[\frac{2-\tilde{\omega}}{2+\tilde{\omega}} \right] \approx \frac{ie^2}{4\pi\hbar} \left[1 + \frac{1}{2} \frac{\tilde{k}^2}{\tilde{\omega}^2} \right] \ln \left[\frac{2-\tilde{\omega}}{2+\tilde{\omega}} \right],$$

from which we read off $\beta_{\text{inter}}^2 = 1/2$.

Aside: Consistency check between Christensen and Guinea paper

The paper by Guinea gives the polarizability, which is related to the conductivity via (see https://hackmd.io/@aligho/ByRI-iJ_ge (https://hackmd.io/@aligho/ByRI-iJ_ge)):

$$\sigma(\mathbf{q}, \omega) = \frac{ie^2\omega}{|\mathbf{q}|^2} P(\mathbf{q}, \omega)$$

Since we are worried only about the imaginary part of the conductivity, we need only worry only about the real part of the polarizability. Therefore, we consider only the contribution from **equation 11** of the paper by Guinea. Furthermore, the Christensen paper looks at the small $|\mathbf{q}|$ limit (e.g. $|\mathbf{q}| \ll \omega/v_F$). In this limit, and taking $\hbar\omega < \varepsilon_F$ (the condition in the Christensen paper), only the first Heaviside function is nonzero. Note also that the function $G(x)$ in the Guinea paper is the same as the function $W(x)$ in the Christensen paper, and the arguments of this function are also the same since:

$$\frac{\hbar\omega + 2\varepsilon_F}{\hbar v_F q} = \frac{\hbar\omega/\varepsilon_F + 2}{q/k_F} = \frac{2 + \tilde{\omega}}{\tilde{q}}$$

Therefore, the polarizability from the Guinea paper gives us a conductivity:

$$\begin{aligned} \sigma(\mathbf{q}, \omega) &= \frac{ie^2\omega}{|\mathbf{q}|^2} \left[-\frac{2\varepsilon_F}{\pi\hbar^2 v_F^2} + \frac{1}{4\pi} \frac{q^2}{\sqrt{(\hbar\omega)^2 - (\hbar v_F q)^2}} \right] \left[W\left(\frac{2 + \tilde{\omega}}{\tilde{q}}\right) - W\left(\frac{2 - \tilde{\omega}}{\tilde{q}}\right) \right] \\ &= \frac{ie^2\tilde{\omega}\varepsilon_F^2/\hbar}{\pi\hbar^2 v_F^2 k_F^2 \tilde{q}^2} \left[-2 + \frac{\hbar^2 v_F^2}{4\varepsilon_F} \frac{\tilde{q}^2 k_F^2}{\sqrt{(\tilde{\omega})^2 \varepsilon_F^2 - (\hbar v_F \tilde{q})^2 k_F^2}} \right] \left[W\left(\frac{2 + \tilde{\omega}}{\tilde{q}}\right) - W\left(\frac{2 - \tilde{\omega}}{\tilde{q}}\right) \right] \end{aligned}$$

By noting that $\varepsilon_F = \hbar v_F k_F$, we simplify the above as:

$$\frac{ie^2\tilde{\omega}}{\pi\hbar\tilde{q}^2} \left[-2 + \frac{1}{4} \frac{\tilde{q}^2}{\sqrt{\tilde{\omega}^2 - \tilde{q}^2}} \right] \left[W\left(\frac{2 + \tilde{\omega}}{\tilde{q}}\right) - W\left(\frac{2 - \tilde{\omega}}{\tilde{q}}\right) \right],$$

which is exactly the same answer as in Thomas's paper. We note, however, that in the very large ω limit (i.e. $\hbar\omega \gg \mu$), the Guinea paper gives a conductivity:

$$\frac{ie^2\tilde{\omega}}{\pi\hbar\tilde{q}^2} \left[-2 + \frac{1}{4} \frac{\tilde{q}^2}{\sqrt{\tilde{\omega}^2 - \tilde{q}^2}} \right] \left[W\left(\frac{\tilde{\omega} + 2}{\tilde{q}}\right) - W\left(\frac{\tilde{\omega} - 2}{\tilde{q}}\right) \right],$$

Appendix 1: Compressibility sum rule in 3D electron gas

From Compressibility (<https://hackmd.io/@aligho/SyE1gAntye>), we know that in the long wavelength, low frequency limit, we must have:

$$\varepsilon(\mathbf{q}) = 1 + \frac{e^2 n^2 \kappa}{\varepsilon_0 |\mathbf{q}|^2} = 1 + \frac{3e^2 n}{2\varepsilon_0 \varepsilon_F |\mathbf{q}|^2} = 1 + \frac{k_{TF}^2}{|\mathbf{q}|^2},$$

where k_{TF} is the Thomas-Fermi screening wavevector (<https://lampz.tugraz.at/~hadley/ss2/ee/screening.php> (<https://lampz.tugraz.at/~hadley/ss2/ee/screening.php>)).

As a sanity check, we note that the above implies that the screened interaction in three dimensions is given by:

$$V(\mathbf{q}) = \frac{e^2}{\varepsilon_0 (|\mathbf{q}|^2 + k_{TF}^2)},$$

which satisfies the Poisson equation (in real space):

$$\nabla^2 V(\mathbf{r}) - k_{TF}^2 V(\mathbf{r}) = -(e^2/\varepsilon_0)\delta(\mathbf{r})$$

Appendix 2: Compressibility sum rule in graphene

From Compressibility (<https://hackmd.io/@aligho/SyE1gAntye>), we know that in the long wavelength, low frequency limit, we must have (the only modification in two dimensions is the potential induced by the modification in charge density):

$$\varepsilon(\mathbf{q}) = 1 + \frac{e^2 n^2 \kappa}{2\varepsilon_0 |\mathbf{q}|},$$

where the compressibility, κ is given by (using the expression for pressure from **Section 4**):

$$\kappa^{-1} = -\Omega \frac{\partial P}{\partial \Omega}, P = \frac{1}{3} \hbar v_F k_F n \rightarrow \kappa^{-1} = \frac{1}{2} \hbar v_F k_F n$$

which gives us:

$$\varepsilon(\mathbf{q}) = 1 + \frac{e^2 n}{\hbar v_F k_F \varepsilon_0 |\mathbf{q}|} = 1 + \frac{e^2 k_F}{\pi \hbar v_F \varepsilon_0 |\mathbf{q}|},$$

Note that this is clearly dimensionless, as required, since $e^2/\varepsilon_0 \hbar v_F$ has units of the fine structure constant, which is dimensionless.

To check our result, we compare with Dynamical polarization of graphene at finite doping (<https://iopscience.iop.org/article/10.1088/1367-2630/8/12/318/pdf>), **equation 13**, from which we get the static dielectric function:

$$\varepsilon(\mathbf{q}) = 1 + \frac{e^2}{2\varepsilon_0 |\mathbf{q}|} \frac{2k_F}{\pi \hbar v_F} = 1 + \frac{e^2 k_F}{\varepsilon_0 \pi \hbar v_F |\mathbf{q}|},$$

which is the same as what was obtained through the compressibility sum rule.

Appendix 3: Effective mass in graphene

We provide an intuitive derivation for the effective mass which appears in the expression for the conductivity. First, we note that the inverse effective mass tensor is given by $m_x^{-1} = \partial_{k_x}^2 (\hbar v_F k) / \hbar^2$, which gives us:

$$m_x^{-1} = \frac{v_F}{\hbar} \frac{\partial^2 k}{\partial k_x^2} = \frac{v_F}{\hbar} \frac{\partial}{\partial k_x} \left[\frac{k_x}{k} \right] = \frac{v_F}{\hbar} \left[\frac{1}{k} - \frac{k_x^2}{k^3} \right] = \frac{v_F}{\hbar k} \sin^2(\theta) \rightarrow$$

$$\left\langle \frac{1}{m_x} \right\rangle = \frac{1}{\pi k_F^2} \frac{v_F}{\hbar} \pi k_F = \frac{v_F}{\hbar k_F} \rightarrow m_{\text{eff}} = \frac{\hbar k_F}{v_F}$$